

Tracking Eye Movements to Extract Visual Patterns for Better Language Problem-Solving Skills in School-aged Children

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Abstract

This study investigated language problem-solving skills and eye-gaze patterns in children with specific language impairment (SLI) compared to typically developing (TD). Additionally, it verified whether visual cues—designed based on the gaze strategies of high-performing children—could enhance task performance. Thirty elementary grades 1-2 students participated in the study. They performed Language Problem-solving Skills Tasks across four domains: situation, cause, emotion, and problem-solution inference. Tasks were administered under two conditions: with and without visual cues that highlighted key information. Performance accuracy and eye-tracking metrics (AOI fixation duration[%], AOI fixation count, and AOI average fixation duration[ms]) were collected and analyzed during both the question and response phases. The findings demonstrate that visual cues serve as effective scaffolding by guiding attention to core information, thereby enhancing language problem-solving skills in children with SLI. This suggests that interventions providing explicit visual guidance can significantly aid the cognitive and linguistic processing of children with language impairments.

CCS Concepts

• **Social and professional topics** → **Children**; • **Human-centered computing** → **User studies**.

Keywords

Specific Language Impaired(SLI); language problem-solving skills; visual cue; scaffolding

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1 Introduction

1.1 Research Background

Language problem-solving skills refer to high-level cognitive-linguistic abilities that involve selecting key cues from a problem, logically linking characters' emotions with the causes and effects of events, and expressing them through language [9, 10, 20, 25]. Problem-solving is defined as a goal-oriented thinking process that encompasses cognitive functions such as attention, memory, inference, and planning, with alternative solution thinking, means-ends thinking, and consequential thinking serving as its core components [20, 28]. Thus, problem-solving requires a series of interconnected processes—generating various solutions, planning procedures, and evaluating outcomes—all of which are closely linked to a child's social behavior and conflict resolution skills [24, 28].

These skills are inextricably linked to linguistic competence, as language plays a pivotal role in the overall understanding and information processing of situational contexts [6, 7, 23]. Consequently, previous research has reported that children with language impairments demonstrate lower performance across all sub-domains of problem-solving. While they may perceive social situations similarly to typically developing (TD) children, contextual understanding is often hindered by the use of maladaptive information processing strategies [1, 13]. Furthermore, deficits in causal reasoning and emotional inference often lead to difficulties in resolving social problems effectively [3, 4, 8, 19, 25, 26, 31, 33]. Despite the established link between language and problem-solving elements, research analyzing these factors through an integrated approach remains limited.

Specific Language Impairment (SLI) is a prevalent condition with an estimated incidence rate of 10–15%, often exerting a negative effect on academic achievement. Children with SLI lack the ability to organize and infer complex situational information despite having normal intelligence, leading to persistent difficulties in key problem-solving processes [1, 8]. These challenges stem not only from linguistic deficits but also from limited information processing capacity and executive function deficits. Notably, children with SLI are highly susceptible to distractor interference, showing a tendency for their gaze to be dispersed rather than selectively attending to key cues in complex visual-linguistic environments [21, 27, 30]. This inefficiency in attention distribution can directly impede language problem-solving, as utilizing various visual cues is essential for situational understanding.



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Eye-tracking data based on the Eye-mind assumption provide valuable information for quantitatively analyzing a child's attention distribution and processing efficiency [12, 14, 15]. Eye-tracking metrics, categorized into temporal and count measures, serve as key indicators for identifying cognitive-linguistic processes in real-time [17, 29, 32]. Previous studies consistently report that children with SLI employ inefficient exploration strategies in tasks requiring the integration of visual and linguistic information, such as vocabulary learning or picture book reading [5, 32]. These findings suggest that when a cognitive load is applied, difficulties in information integration lead to decreased task performance, highlighting eye-tracking as a crucial tool for revealing the underlying mechanisms of children's problem-solving skill [11].

1.2 Research Question

While traditional assessments focus on accuracy, they fail to capture the real-time cognitive mechanisms of complex language problem-solving. Eye-tracking addresses this gap by revealing underlying processing inefficiencies, particularly in children with Specific Language Impairment (SLI). Thus, we investigate: (1) How do visual cues affect task performance across task sub-domains (situation, cause, emotion, problem-solution) for TD and SLI groups? (2) How do visual cues influence eye-tracking patterns between the two groups during the question and response phases?

2 Method

2.1 Research Design

2.1.1 Participants. A total of 30 children in the 1st or 2nd grade of elementary school participated in this study, consisting of 11 children with Specific Language Impairment (SLI) and 19 typically developing (TD) children. Inclusion criteria for the SLI group were established based on Leonard (1998) [18] as follows: (1) Normal hearing and social-emotional development as reported by parents and teachers; (2) A non-verbal IQ score of 85 (1 SD) or higher on the Kaufman Brief Intelligence Test-II (KBIT-II; [22]); (3) Scores below the 10%ile in either receptive or expressive vocabulary on the Receptive and Expressive Vocabulary Test (REVT; [16]); (4) No history of behavioral, emotional, sensory, or neurological disorders. Statistical analyses confirmed that there were no significant differences between the two groups in chronological age and non-verbal IQ. However, the TD group showed significantly higher scores in both expressive and receptive vocabulary compared to the SLI group.

2.1.2 Experimental materials. Visual cues were developed based on pilot study eye-tracking data from four 5th-grade students (two out of six initially recruited were excluded for failing to reach 90% accuracy), as adult gaze strategies might present an excessive cognitive gap for Grade 1–2 students to emulate. This ensured that the visual scaffolds were grounded in a matured yet attainable level of linguistic reasoning within a shared educational context. Using an SMI stationary eye-tracker, heatmaps generated during task performance were analyzed to identify common focal points and objects, which were subsequently designated as visual cues. This process resulted in the selection of three cues for Situation Inference and two cues each for Cause, Emotion, and Problem-solution Inference. To ensure stimulus consistency, these cues occupied less than 20% of

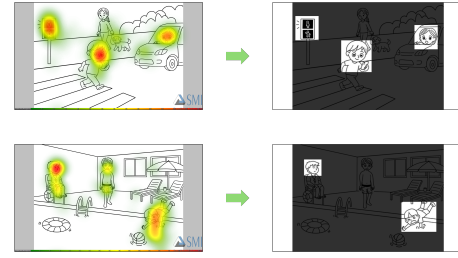


Figure 1: Visual Cue Extraction Process: Heatmaps obtained from the pilot study (Left) and visual cues extracted for the main experiment (Right)

the total area and were highlighted by reducing non-target brightness by 80%. Stimuli were presented in a randomized order in the main experiment. An example of the stimulus containing these visual cues is illustrated in Figure 1.

2.1.3 Procedure. To measure children's language problem-solving skills, this study utilized a task adapted and supplemented from the Test of Problem Solving (TOPS; [2]). The task consisted of 20 black-and-white line drawings depicting everyday conflict situations resulting from carelessness or misjudgment, accompanied by a total of 80 questions. These questions were categorized into four domains: Situation, Cause, Emotion, and Problem-solution Inference (20 items per domain), with validity confirmed through review by six language pathologists. The stimulus set was divided into two conditions—with and without visual cues—containing 10 situational contexts each. The non-visual cue condition was presented first to prevent potential learning effects; this design choice ensured that participants' performance in the baseline condition was not influenced by prior exposure to the gaze-informed scaffolds. Sample questions for the Language Problem-solving Task are presented in Table 1.

For real-time collection of eye-tracking data, the task was designed using the SMI Experiment 3.5 software, and data were recorded using an SMI REDn Scientific stationary binocular eye-tracker (60Hz sampling rate, 0.4° accuracy). The device was attached to the bottom of a 24-inch monitor, and participants performed the task while maintaining a distance of 60–70 cm from the screen. Before the experiment, a 5-point calibration was conducted, and the task was performed only when an accuracy level of $\geq .7$ was achieved.

Before the main experiment, a practice session was provided to ensure comprehension of the tasks. The experiment followed a fixed order, starting with the none-visual cue condition followed by the visual cue condition. To eliminate the effects of difficulty variance between item sets, counterbalancing was applied across participants. Each item presented a visual stimulus simultaneously with a recorded audio question. The researcher provided no feedback on the participants' responses. While questions were generally presented once, they were repeated upon request. If no response was given within one minute, the item was marked as "no response," and the next trial commenced. The overall experimental procedure is illustrated in Figure 2.

Domains	Questions
Situation Inference	What happened in this situation?
Cause Inference	<i>Why</i> - question (ex: Why did this event occur?)
Emotion Inference	How do you think [Character's name] feels?
Problem-solution Inference	<i>How</i> - question (ex: How can [Character's name] solve this problem?)

Table 1: Sample Questions by Domain

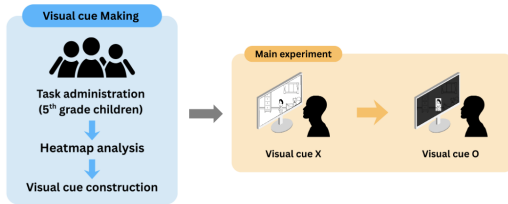


Figure 2: Overview of the Experimental Procedure

2.1.4 Data analysis. Language problem-solving skills was assessed by transcribing and scoring participants' verbal responses, including self-corrections. A 3-point scale (0–2) was applied based on the appropriateness and logical completeness of the answers: 2 points for adequate and sufficient inferences, 1 point for accurate but poorly delivered or insufficient information, and 0 points for incorrect or no response. The maximum raw score for each domain was 20 points. To ensure consistency, inter-rater reliability was analyzed using 10% of the data re-scored by the research assistants, yielding a high reliability coefficient of $r = .90$.

Eye-tracking data was analyzed using SMI Be Gaze 3.7 software, with only data showing a tracking rate above 70% included in the final analysis [12]. The analysis was divided into two phases: the 'Question Phase', involving auditory-visual integration, and the 'Response Phase', involving speech planning and production. Areas of Interest (AOIs) were defined as the entire visual cue area in the Visual Cue condition, and as key objects or regions in the No Visual Cue condition. In both conditions, AOI coverage was controlled to remain under 30% of the total screen area. Dependent variables included AOI fixation duration(%), AOI fixation count, and AOI average fixation duration(ms), which serve as indicators of information processing efficiency and attentional focus.

3 Results

3.1 Impact of Visual Cues on Language Problem-Solving Performance

The results indicate that visual cues play a critical role in enhancing language problem-solving skills, particularly for children with SLI. A three-way mixed ANOVA confirmed significant main effects for both group ($F(1,28)=21.633, p<.001, \eta p^2=.436$) and visual cues ($F(1,28)=32.902, p<.001, \eta p^2=.540$). Overall, TD children outperformed children with SLI, and performance in the visual cue condition was significantly higher than in the non-visual condition across both groups.

Crucially, a significant interaction effect was observed between group and visual cues ($F(1,28)=19.564, p<.001, \eta p^2=.411$), indicating

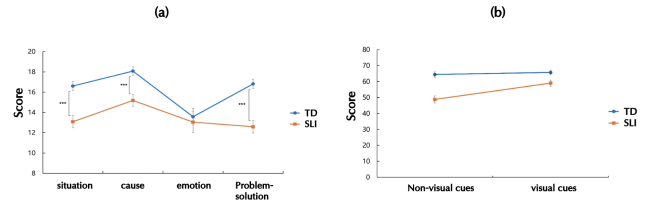


Figure 3: Language problem-solving skills scores by group according to (a) task sub-domains and (b) visual cue conditions

that the benefit of visual cues support differed by group. Post-hoc analysis revealed that while TD children showed no significant performance difference based on the presence of visual cues ($p=.288$), children with SLI exhibited a significant improvement when visual cues were provided ($p<.001$). This suggests that visual cues serve as an effective compensatory mechanism for the SLI group.

Regarding specific domains, the analysis revealed a significant interaction between task sub-domain and group ($F(1.56,43.56)=3.849, p=.039, \eta p^2=.121$). While TD children scored significantly higher than SLI children in Situation, Cause, and Problem-solution inferences ($p<.001$), no significant difference was found between the two groups in the Emotion Inference domain ($p=.689$). Furthermore, visual cues significantly improved performance in all domains except for Emotion Inference ($p=.231$), suggesting that emotional processing may rely on processing mechanisms distinct from other domains. The results are illustrated in Figures 3.

3.2 Effect of Visual Cues on Gaze Patterns During the Question Phase

Eye-tracking data during the question phase revealed that visual cues support significantly engaged participants across both groups. A three-way mixed ANOVA on fixation duration ratio showed a significant main effect of visual cues ($F(1,28)=146.29, p<.001, \eta p^2=.839$), with participants looking at the screen significantly longer in the visual condition ($M=59.28$) compared to the non-visual condition ($M=40.51$). There was no significant group difference, indicating that children with SLI attended to visual cues as actively as TD children while processing the questions.

Regarding fixation counts, a similar pattern emerged. Participants demonstrated significantly higher fixation counts in the visual condition ($M=6.60$) than in the non-visual condition ($M=4.39$; $F(1,28)=210.39, p<.001, \eta p^2=.883$). A significant main effect of task domain was also found ($F(2.04,57.15)=117.38, p<.001, \eta p^2=.807$). Notably, the Problem-solution Inference domain elicited the highest number of fixations ($M=7.42$), suggesting that this domain required

more intensive visual scanning to gather relevant information compared to other domains.

Furthermore, a significant interaction between domain and visual cues on fixation counts ($F(1.91,53.41)=13.12$, $p<.001$, $\eta p^2=.319$) indicated that the increase in visual scanning due to visual cues was the most pronounced in the Problem-solution Inference domain. This highlights the critical role of visual information in supporting the cognitive processing required for complex Language Problem-solving Tasks.

3.3 Effect of Visual Cues on Gaze Patterns During the Response Phase

In the response phase, where participants formulated and articulated their answers, visual cues continued to play a supportive role. Analysis of the fixation duration revealed a significant main effect of visual cues ($F(1,28)=27.71$, $p<.001$, $\eta p^2=.497$), with both groups spending a significantly larger proportion of time fixing their gaze on visual stimuli in the visual cue condition ($M=39.43$) compared to the non-visual cue condition ($M=32.43$).

However, distinct processing strategies emerged between groups regarding fixation counts, particularly in complex tasks. A significant interaction between task sub-domain and group was found ($F(1.86,51.97)=3.61$, $p=.037$, $\eta p^2=.114$). Post-hoc analysis indicated that in the Problem-solution Inference domain, TD children demonstrated significantly higher fixation counts than children with SLI. This suggests that TD children actively re-visited visual cues to verify information while answering complex questions, whereas children with SLI may lack this strategic retrieval mechanism.

Additionally, an interaction between task sub-domain and visual cues ($F(1.99,55.65)=4.08$, $p=.022$, $\eta p^2=.127$) confirmed that the presence of visual cues significantly increased fixation counts specifically in the Situation and Problem-solution Inference domains. No significant effects were found for average fixation duration, implying that the differences in visual processing were driven by the frequency of looks rather than the duration of individual gazes.

4 Conclusion and Implications

Our findings confirm that visual cues act as a vital compensatory mechanism for children with SLI, significantly narrowing the performance gap with TD peers, particularly in the Cause Inference domain. However, the Emotion Inference domain proved challenging for both groups even with the presence of visual cues; it required integrating causal contexts with limited non-verbal information, suggesting that simple spatial scaffolding may not suffice for high-level emotional reasoning. While visual cues enhanced information processing efficiency for both groups, eye-tracking analysis revealed a critical strategic divergence during the response phase: TD children actively re-visited visual stimuli to verify information, whereas children with SLI struggled to sustain attention, likely due to limited working memory capacity which hinders the continuous integration of visual information during verbal production. These findings suggest that even when SLI children provide correct answers, their underlying processing may remain inefficient, highlighting the limitations of relying solely on outcome-based assessment. Translating these findings into HCI design, we propose two key guidelines for educational technologies. First, systems

should offer adaptive visual scaffolding. Since static images alone do not trigger strategic checking in children with SLI, interactive interfaces should mimic TD strategies by dynamically highlighting or zooming in on critical regions when inefficient gaze patterns are detected. Second, future tools must integrate process-based analytics. By leveraging gaze tracking or interaction logs, systems can evaluate how a child arrives at an answer, providing real-time feedback on the problem-solving strategy rather than just the final outcome. While limited by a small sample size and a fixed task sequence—chosen to prevent baseline bias—these findings lay a critical foundation for future longitudinal research on adaptive scaffolding.

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